ClimateCover Australia

Insurance Affordability and Property Risk Intelligence Platform

A professional project handover covering the business problem, solution design, data sources, methodology, code structure, current results and deployment path.

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1. Executive Summary

ClimateCover Australia is a regional risk intelligence platform that identifies where climate-related property exposure overlaps with household insurance affordability pressure. It is designed for insurers, banks, government resilience teams, regulators and consulting stakeholders who need a clear, explainable view of regional vulnerability.

2,353	29.88	4.26%	21	18
SA2 regions	Avg property risk	Avg premium to income	Severe affordability	High or severe risk

- Uses official ABS SEIFA 2021 SA2 records for regional names, population and socio-economic indicators.
- Uses real ABS Census 2021 SA2 household indicators for income, mortgage repayments, rent and household count.
- Creates calculated affordability and property risk metrics for decision support.
- Includes a Streamlit dashboard with executive overview, risk explorer, region profile, methodology and data quality pages.
- Includes an automated refresh workflow that can rebuild public data and source metadata from GitHub Actions.

2. Business Problem and Value

Australian households face increasing home insurance affordability pressure as natural hazards, rebuilding costs and income vulnerability interact. Organisations need to understand where risk is concentrated, where affordability stress is emerging and where resilience investment may have the greatest impact.

Stakeholder	Problem They Need To Solve	How ClimateCover Helps
Insurers	Identify regional exposure and affordability pressure without relying only on portfolio-level summaries.	Ranks regions by property risk, premium burden and intervention priority.
Banks	Understand climate and insurance stress that may affect household resilience and mortgage risk.	Combines household income, mortgage pressure, risk scores and affordability bands.
Government	Prioritise mitigation, resilience funding and community support.	Highlights where physical risk overlaps with household vulnerability.
Consultants	Demonstrate data platform, analytics, AI-style briefing and solution design capability.	Shows a complete data-to-dashboard architecture with explainable scoring.

3. What Was Built

Layer	What It Does	Key Files / Tables
Public data acquisition	Downloads and prepares ABS SEIFA and ABS Census SA2 household data.	scripts/prepare_real_abs_data.py; data/raw/
Foundation model	Creates the regional spine, SEIFA facts and demographic facts.	dim_region; fact_seifa; fact_demographics; foundation_region_profile
Climate and hazard layer	Adds complete climate and hazard indicators for every region.	fact_climate; fact_hazard; foundation_region_climate
Risk and affordability layer	Calculates premium burden, affordability bands, risk scores and intervention priority.	fact_affordability; fact_property_risk; foundation_region_risk
Dashboard application	Presents executive KPIs, exploration tables, regional profiles, methodology and data quality.	app/Home.py; app/pages/*; src/dashboard/
Automation	Refreshes public data and updates source metadata on a schedule.	.github/workflows/refresh-public-data.yml; scripts/refresh_public_data.py

The result is a working local analytics product. Python and Pandas prepare the data, DuckDB stores the analytical model, and Streamlit presents decision-ready pages. The same repository can be deployed to Streamlit Community Cloud.

4. Data Sources and Lineage

The platform separates data into real public data, modelled indicators and calculated metrics. This is important because it makes the dashboard honest and explainable while still keeping the product usable end to end.

Layer	Fields	Source	Status	Business Use
Household Indicators	Income, rent, mortgage, household count	ABS Census 2021 General Community Profile SA2 DataPack	Real Public Data	Insurance affordability denominator and household scale
Regional Reference	SA2 code, SA2 name, state, population	ABS SEIFA 2021 SA2 workbook	Real Public Data	Regional coverage and joins
Socio-economic Indicators	IRSD and IER scores and deciles	ABS SEIFA 2021 SA2 workbook	Real Public Data	Community vulnerability and affordability context
Insurance Affordability	Estimated annual premium and premium-to-income ratio	Explainable affordability model	Calculated Metric	Affordability stress screening
Property Risk	Property risk score, risk band, intervention priority	Explainable weighted scoring model	Calculated Metric	Prioritisation and executive decision support
Climate Indicators	Rainfall, temperature, heat days, rainfall anomaly	BOM-style prepared indicators	Modelled Indicator	Climate pressure signal
Hazard Indicators	Flood, bushfire, cyclone, storm scores	Geoscience Australia, data.gov.au and state open-data hazard layers	Modelled Indicator	Physical property risk exposure

Current real public data includes ABS SEIFA 2021 and ABS Census 2021 General Community Profile SA2 household fields. Climate and hazard fields are currently modelled indicators pending prepared BOM, Geoscience Australia and state hazard extracts. Insurance premiums are modelled affordability estimates because public national SA2-level insurer quote data is not openly available.

5. Current Results

The current database contains 2,353 Australian SA2 regions. The table below lists the highest intervention priority regions in the latest build.

Region	State	Risk	Risk Band	Premium / Income	Affordability	Priority
Victoria River	Northern Territory	59.50	High	9.82%	Severe	75.70
Riverview	Queensland	56.17	High	9.60%	Severe	73.70
Tiwi Islands	Northern Territory	54.63	High	13.24%	Severe	72.78
Southern Moreton Bay Islands	Queensland	51.29	High	13.06%	Severe	70.78
Deception Bay	Queensland	51.29	High	8.39%	Severe	70.78
Torres Strait Islands	Queensland	51.29	High	10.17%	Severe	70.78
Palm Island	Queensland	51.29	High	10.83%	Severe	70.78
Lakes Creek	Queensland	55.17	High	7.79%	High	70.53
Southport - North	Queensland	50.29	High	8.54%	Severe	70.18
Pialba - Eli Waters	Queensland	50.29	High	9.98%	Severe	70.18

6. Methodology

Property Risk Score

Component	Weight	Purpose
Flood risk	30%	Captures flood exposure contribution.
Bushfire risk	25%	Captures bushfire exposure contribution.
Cyclone and storm risk	15%	Captures severe weather exposure.
Climate indicators	10%	Adds rainfall, heat and temperature pressure.
Historical hazard indicator	10%	Represents combined hazard signal until event history is added.
Socio-economic vulnerability	10%	Adds household and SEIFA vulnerability context.

Affordability Methodology

- Annual household income is calculated from median weekly household income multiplied by 52.
- Premium-to-income percentage compares the estimated annual insurance cost with annual household income.
- Bands are Low under 2%, Moderate from 2% to under 4%, High from 4% to under 8.33%, and Severe at 8.33% or above.
- The Severe threshold approximates one month of annual household income.

7. Dashboard Pages

Page	Purpose	Key Features
Home	Introduces the product and data status.	Business problem, solution overview, source list, navigation.
Executive Overview	Gives a national decision view.	KPI cards, risk distribution, affordability distribution, top ranked regions, generated executive briefing.
Risk Explorer	Lets users filter and compare regions.	State, risk band and affordability filters; searchable table; CSV export.
Region Profile	Explains a selected region.	Demographics, SEIFA, climate, hazard scores, affordability, generated regional briefing.
Methodology	Explains the scoring and governance model.	Public data foundation, affordability model, risk scoring, assumptions and limitations.
Data Quality	Shows source credibility and validation status.	Lineage table, refresh status, null checks, duplicate checks, range checks.

8. Code Structure

Path	Role
app/	Streamlit multi-page application.
app/pages/	Executive overview, risk explorer, region profile, methodology and data quality pages.
src/dashboard/	Reusable dashboard queries, chart builders, formatting and generated insight text.
src/data_ingestion/	Reusable ingestion utilities for regions, SEIFA, demographics, climate, hazard and insurance estimates.
src/transformation/	Builds foundation, climate and risk analytical models.
src/utils/	DuckDB helpers and validation functions.
scripts/	Pipeline runners, ABS preparation, database build and public data refresh.
docs/	Data sources, data dictionary, methodology, manifest and source catalog.
db/	Local DuckDB database.
data/raw/	Downloaded and prepared source extracts.
output/pdf/	Generated PDF documentation.

9. Automation and Deployment

- Local app launch command: streamlit run app/Home.py
- Local refresh command: python scripts/refresh_public_data.py
- GitHub Actions workflow: .github/workflows/refresh-public-data.yml
- The refresh workflow prepares public data, rebuilds the DuckDB model for validation, updates the refresh manifest and commits changed source metadata.
- The app can bootstrap the database when the required decision layer is missing.

10. Limitations and Next Steps

- Climate indicators should be replaced with prepared BOM SA2 climate features.
- Hazard indicators should be replaced with Geoscience Australia and state spatial hazard layers.
- Insurance premiums remain modelled affordability estimates because public national SA2-level insurer premium data is not openly available.
- A production implementation should add automated tests for scoring and data quality thresholds.
- A client-grade implementation could connect insurer portfolio, claims, property and mitigation data under appropriate governance.

11. Portfolio Value

This project demonstrates business analysis, enterprise data modelling, public data ingestion, data quality governance, explainable analytics, AI-style generated summaries, dashboard product design and deployment readiness. It is positioned for Australian data, AI, analytics and consulting roles where employers want to see practical solution design rather than a classroom notebook.